



CIKM 2024 Notification for Paper 712

From Full Research Paper Track at CIKM 2024 <CIKM2024-pc@easychair.org>

Date Tue 16/07/2024 17:52

To Kieu Hai Dang (CECS) <dang.kh@vinuni.edu.vn>

Dear Dang,

We regret to inform you that your paper 712 entitled Keyword-driven Retrieval-Augmented Large Language Models for Cold-start User Recommendations was not accepted for the Full Research Paper track at the upcoming 33rd ACM International Conference on Information and Knowledge Management (CIKM 2024).

For the Full Research Paper track, we received 1531 complete submissions. After a thorough vetting process, 1496 contributions were reviewed. Based on a thorough and rigorous review process we have accepted 347 papers. This yields an overall acceptance rate of 23%.

This year, the Full Research Paper track has been very competitive, with very high-quality submissions. Each paper received at least three reviews and was assigned one Senior PC member, who led discussions on the merits and weaknesses of each submission and gave a final recommendation. Based on the reviews, the SPC recommendations, and our analysis of the contributions, we curated a set of accepted papers showcasing the state-of-the-art in the different areas of interest for the CIKM community.

This was a very competitive process, so please do not be discouraged by the rejection. We include below the reviews for your submission and hope that the comments from the reviewers can help improve your work. Note that ACM CIKM does not have a rebuttal period and therefore decisions are final.

We also would like to take this opportunity to remind you that there are several workshops co-located with CIKM, all with deadlines in early August that might be of interest to you, as another opportunity to share your valuable work and elicit further feedback from the CIKM community. For more information, see <https://cikm2024.org/workshops/>

Best regards,

Carlotta, Zhifeng, and Sole
Full Research Papers Track Chairs
cikm2024-pc@easychair.org

SUBMISSION: 712

TITLE: Keyword-driven Retrieval-Augmented Large Language Models for Cold-start User Recommendations

----- METAREVIEW -----

The paper introduces an innovative framework addressing the cold-start problem in recommender systems and presents a clear and accessible methodology. However, it is limited by the below issues.

1. The evaluation is narrow, focusing on a single dataset and lacking generalizability, and there is a notable absence of user studies to validate the real-world effectiveness.
2. Did not compare with state-of-the-art cold-start recommendation techniques.
3. Did not thoroughly explore different keyword extraction methods or potential biases in LLM rankings.

----- REVIEW 1 -----

SUBMISSION: 712

TITLE: Keyword-driven Retrieval-Augmented Large Language Models for Cold-start User Recommendations

AUTHORS: Dang Kieu, Minh Duc Nguyen, Thanh-Son Nguyen and Dung D. Le

----- Overall recommendation -----

SCORE: 1 (weak accept)

----- Relevance to CIKM -----

SCORE: 4 (good)

----- Originality of the Work -----

SCORE: 3 (fair)

----- Technical Soundness -----

SCORE: 4 (good)

----- Quality of Presentation -----

SCORE: 4 (good)

----- Impact of Ideas or Results -----

SCORE: 3 (fair)

----- Reproducibility of Methods -----

SCORE: 1 (very poor)

----- Detailed Comments to the Author(s) -----

Summary Of Strengths:

1. The paper introduces KALM4Rec, a novel framework designed to tackle the cold-start recommendation problem using keyword-driven retrieval and LLM-based re-ranking. This is a good contribution, as the cold-start problem is a well-known challenge in recommender systems.
2. The paper excels in its clear presentation, particularly through its use of illustrative images. These visuals effectively demystify the complex methodology and operational flow of the KALM4Rec framework, making the paper more accessible and understandable to readers.

Summary Of Weakness:

1. It is difficult to demonstrate the scalability of the approach proposed in this paper by doing experiments on only one dataset.
2. The annotation is a little bit confusing in section 3, which makes it harder to follow. For example, the meaning of Ds is not explained.
3. Discussion. The performance of LLMs in re-ranking candidates can be affected by the order of keywords and examples in prompts. I think the observation and subsequent discussion here about the impact of the order of keywords and examples would be valuable to readers and should also be incorporated during revision.
4. It is recommended that the authors include standard deviation in the table of experimental results to increase the credibility and interpretability of the results.

----- Summary to support your recommendation -----

The KALM4REC framework offers a novel approach to improving recommender systems by leveraging large language models (LLMs) and keyword extraction, but it has several limitations. Only one dataset was used, which is not enough to demonstrate the scalability of the methodology. Additionally, lack of discussion of keyword order. Further research is needed to address these issues and enhance the framework's scalability and effectiveness.

----- REVIEW 2 -----

SUBMISSION: 712

TITLE: Keyword-driven Retrieval-Augmented Large Language Models for Cold-start User Recommendations

AUTHORS: Dang Kieu, Minh Duc Nguyen, Thanh-Son Nguyen and Dung D. Le

----- Overall recommendation -----

SCORE: -1 (weak reject)

----- Relevance to CIKM -----

SCORE: 4 (good)

----- Originality of the Work -----

SCORE: 3 (fair)

----- Technical Soundness -----

SCORE: 4 (good)

----- Quality of Presentation -----

SCORE: 3 (fair)

----- Impact of Ideas or Results -----

SCORE: 4 (good)

----- Reproducibility of Methods -----

SCORE: 3 (fair)

----- Detailed Comments to the Author(s) -----

Strong Points:

1. The paper presents KALM4Rec, an approach combining keyword-driven retrieval and LLM-based re-ranking for cold-start user recommendations. While individual components (keyword extraction, retrieval models, LLM prompting) are based on existing techniques, combining them for cold-start recommendations is novel and aligns with current research trends.
2. The work provides a comprehensive comparison of different LLM models (Gemini, GPT-3.5, Mistral 8B, Llama3 8B) and prompting strategies (zero-shot, few-shot) for recsys tasks. This analysis offers insights into the relative strengths of these models and approaches in the context of recsys tasks, which could be useful for practitioners in the field.
3. The paper considers efficiency well, demonstrating that using keywords instead of full reviews significantly reduces token usage in LLM prompts while maintaining or improving performance.

Weak Points:

1. The evaluation is conducted only on restaurant recommendations from three English-speaking cities, which may limit the generalizability of the findings. The paper also lacks a user study to validate the real-world effectiveness and user satisfaction with the recommendations generated by KALM4Rec.
2. While the paper compares with some baselines, it doesn't include comparisons with recent advanced cold-start recommendation techniques.
3. The paper doesn't thoroughly examine different keyword extraction techniques and their impact on recommendation quality.
4. The examination of potential biases in LLM rankings is relatively brief and the work could benefit from a more in-depth investigation.
5. Presentation Nits:
 - (a) Section 1: "obtain most possible restaurants for cold-start users" seems should be most suitable restaurants or best possible restaurants?
 - (b) Section 3.2.1: "For CBR, profile of all users includes cold-start user is presented" -- including cold-start users?

----- Summary to support your recommendation -----

The paper introduces KALM4Rec, a promising approach that combines keyword-driven retrieval and LLM-based re-ranking for cold-start user recommendations, potentially impactful for recommender systems. It provides a comprehensive comparison of different LLM models and prompting strategies, offering useful insights for practitioners and demonstrates efficiency in reducing token usage by using keywords.

However, the evaluation's focus on restaurant recommendations from three English-speaking cities limits generalizability and the lack of a user study diminishes real-world effectiveness validation. The work also omits comparisons with recent advanced cold-start techniques, provides a very brief examination of potential LLM biases and lacks a thorough analysis of different keyword extraction methods.

----- REVIEW 3 -----

SUBMISSION: 712

TITLE: Keyword-driven Retrieval-Augmented Large Language Models for Cold-start User Recommendations

AUTHORS: Dang Kieu, Minh Duc Nguyen, Thanh-Son Nguyen and Dung D. Le

----- Overall recommendation -----

SCORE: -1 (weak reject)

----- Relevance to CIKM -----

SCORE: 4 (good)

----- Originality of the Work -----

SCORE: 3 (fair)

----- Technical Soundness -----

SCORE: 3 (fair)

----- Quality of Presentation -----

SCORE: 3 (fair)

----- Impact of Ideas or Results -----

SCORE: 3 (fair)

----- Reproducibility of Methods -----

SCORE: 4 (good)

----- Detailed Comments to the Author(s) -----

1. summary

In this paper, author proposes a novel framework, called KALM4Rec (Keyword-driven Retrieval-Augmented Large Language Models for Cold-start User Recommendations), which leverages keyword-driven retrieval model and LLMs for the limitations of input length, hallucination and re-ranking purposes, in order to solve cold-start user appeared in RS. This work also conducts extensive experiment to demonstrate effectiveness of KALM4Rec framework in cold-start users recommendation.

2. weak point

- It seems like more contributions should be in the paper.
- There are some typos in the paper that author should be more careful in writing paper.
- It is better for this work adding more innovative elements.

3. detailed

- author should also demonstrate re-ranking LLM model will not affect conventional users while it benefits cold-start users.
- This work only collects data in restaurant review, does it work well in other domain?
- there is a typo in related work section. "... that arises when dealing when(with) new users ..."
- There were some works using LLMs to address recommendations for cold-start user, also mentioned in related work.
- there is a same symbol appeared in equation2 and equation3.
- In figure 4, the description(title) is different from the value in left part figure, 1 or 0, and what does it mean?
- author should conducts more experiments with recommendations for cold-start users which is the key point in this paper. How many cold-start users are in the total datasets?
- Is this typo in equation 6 with symbol "swr"?

----- Summary to support your recommendation -----

This paper well defined problems in the recommendation system field with cold-start users.

Also, it has a good structure for the audiences to easily understand the work.

However, this work should add more contributions with novel approaches, as well as experiments in various aspects with cold-start users.